Population Projections & Demographic Forecasting

Lecture 3: Matrix Projections II

Ugofilippo Basellini
basellini@demogr.mpg.de
Max Planck Institute for Demographic Research

April 29, 2026

Course overview

- Lecture 1: Population Projections & Cohort Component Method
- Lecture 2: Matrix Projections I
- Lecture 3: Matrix Projections II
- Lecture 4: Demographic Forecasting & Lee-Carter

Small recap

In Lecture 2, we have seen the advantages of using the matrix formulation of the cohort component method for population projections

- can significantly speed up your code and computations
- can incorporate female and male populations within the same setting

$\Rightarrow$ what about migration and time-specific assumptions on future demographic components?

Projection of an open population

- Let $M[0,T]$ be the net migrations between 0 and T. Then $N(T) = N(0) + B[0,T] - D[0,T] + M[0,T]$ (balancing equation of population growth)
- Dividing by number of person-years lived:

$$\begin{aligned} \frac{N(T) - N(0)}{PY[0,T]} &= \frac{B[0,T]}{PY[0,T]} - \frac{D[0,T]}{PY[0,T]} + \frac{M[0,T]}{PY[0,T]} \\ r &= b - d + m \end{aligned}$$

- crude growth rate r expressed in terms of the three demographic components, i.e. birth (b), death (d) and net migration (m) crude rates
- r is used, for example, in constant exponential growth model $N(T) = N(0)e^{rT}$

Requiem for the net migrant (Rogers 1990)

- the net migrant is “a nonexistent category of individuals”
- net migration rate is a flawed measure: it is the difference between a true rate (out-migration) and a prevalence (in-migration). Denominators are different!
- they obscure regularities in age profiles of migration

$\Rightarrow$ Use net migration counts instead

CCM with an open population

- Formal difficulty of integrating migration is that migration continuously affects the population at risk of dying and giving birth
- Convenient approach: divide number of migrants into two discrete quantities, and assume that half move at the start of the interval, and half at the end of it
- Let $I_x[t, t+5]$ be the net flow if immigrants (it can be negative if migration balance is negative). Then:

$$N_{x+5}(t+5) = \left(N_x(t) + \frac{I_x[t, t+5]}{2}\right) s_x + \frac{I_{x+5}[t, t+5]}{2}$$

- In matrix notation: $\boldsymbol{N}(t+5) = \boldsymbol{L}[t, t+5]\left(\boldsymbol{N}(t) + \frac{1}{2}\boldsymbol{I}(t)\right) + \frac{1}{2}\boldsymbol{I}(t)$

Derivation of net migration counts

- Can be computed as the residual part of the balancing equation of population growth - these tend to show erratic year-to-year fluctuations
- alternatively, we can use a parametric model, such as the Rogers and Castro (1981) model migration schedules
- only need the total number of net migrants, which is distributed over the age groups using parametric functions

Rogers and Castro (1981) migration schedule

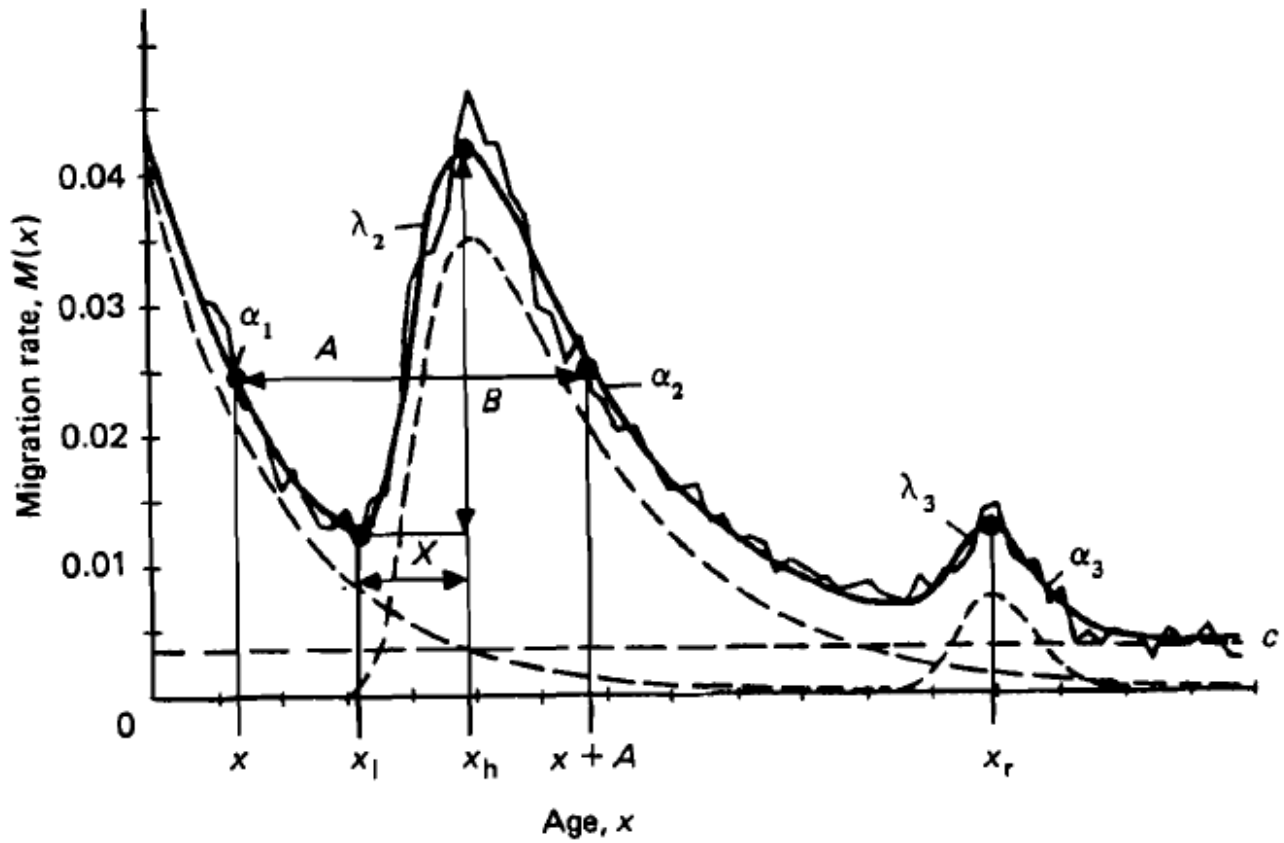


FIGURE 4 The model migration schedule.

Four components: (i) a negative exponential curve of the *pre-labor force* ages; (ii) a left-skewed unimodal curve of the *labor force* ages; (iii) an almost bell-shaped curve of the *post-labor force* ages; and (iv) a constant curve

Rogers and Castro (1981) migration schedule

- let's focus on first two components of migration and the constant, i.e.

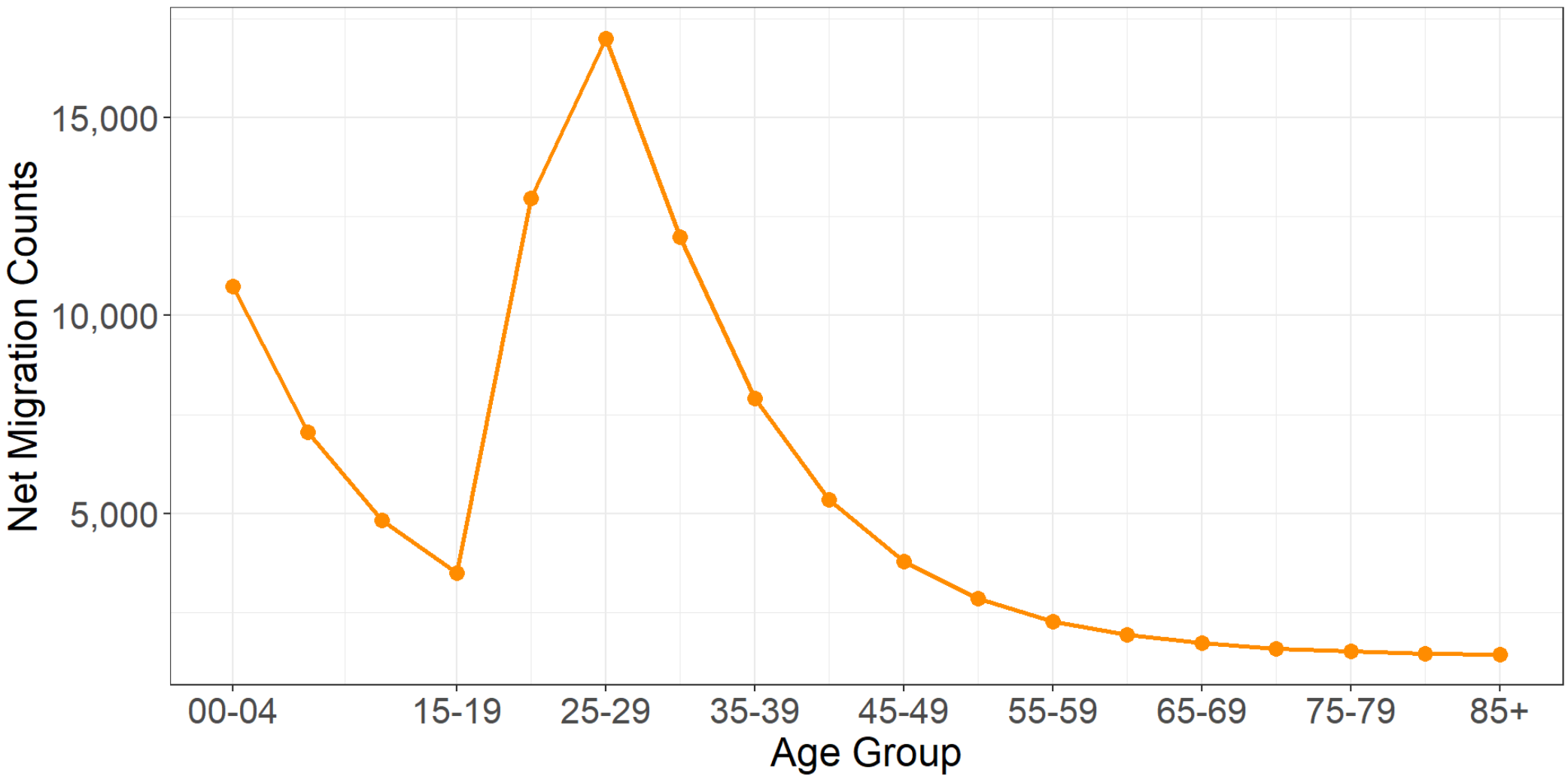
$$m(x) = a_1 \exp\left(-\alpha_1 x\right) + a_2 \exp\left\{-\alpha_2\left(x-\mu_2\right) - \exp\left[-\lambda_2\left(x-\mu_2\right)\right]\right\} + c$$

- the fundamental RC parameters are provided by the `rc_model_fund` in the `migest` package (Abel 2021)
- obtain age-specific counts by multiplying total net migration by the RC schedule

RC migration schedule in R

```r
### cleaning the workspace
rm(list=ls())
## loading packages
library(tidyverse)
library(migest)
## check the fundamental RC parameters
rc_model_fund
my.pars <- rc_model_fund %>%
  select(value) %>% pull()
## function to construct RC schedule
mxRG <- function(x,pars){
  t1 <- pars[1]*exp(-pars[2]*x)
  t2 <- pars[3]*exp(-pars[4]*(x-pars[5])-exp(-pars[6]*(x-pars[5])))
  mx <- t1+t2+pars[7]
  mx <- mx/sum(mx)
  return(mx)
}
## five-year age groups (works well also with one-year)
x <- seq(0,85,5)
```

RC migration schedule in R



Including net migration: Exercise

Exercise

Load yesterday's data `EDSD.lecture3.Rdata`. Project the **female** population for 20 periods ahead assuming an overall amount of 25000 net migrants in each period. Compare your projections with and without the migration component.

Hint: adapt your population projection function by including the net migration counts (by age, derived from the RC schedule).

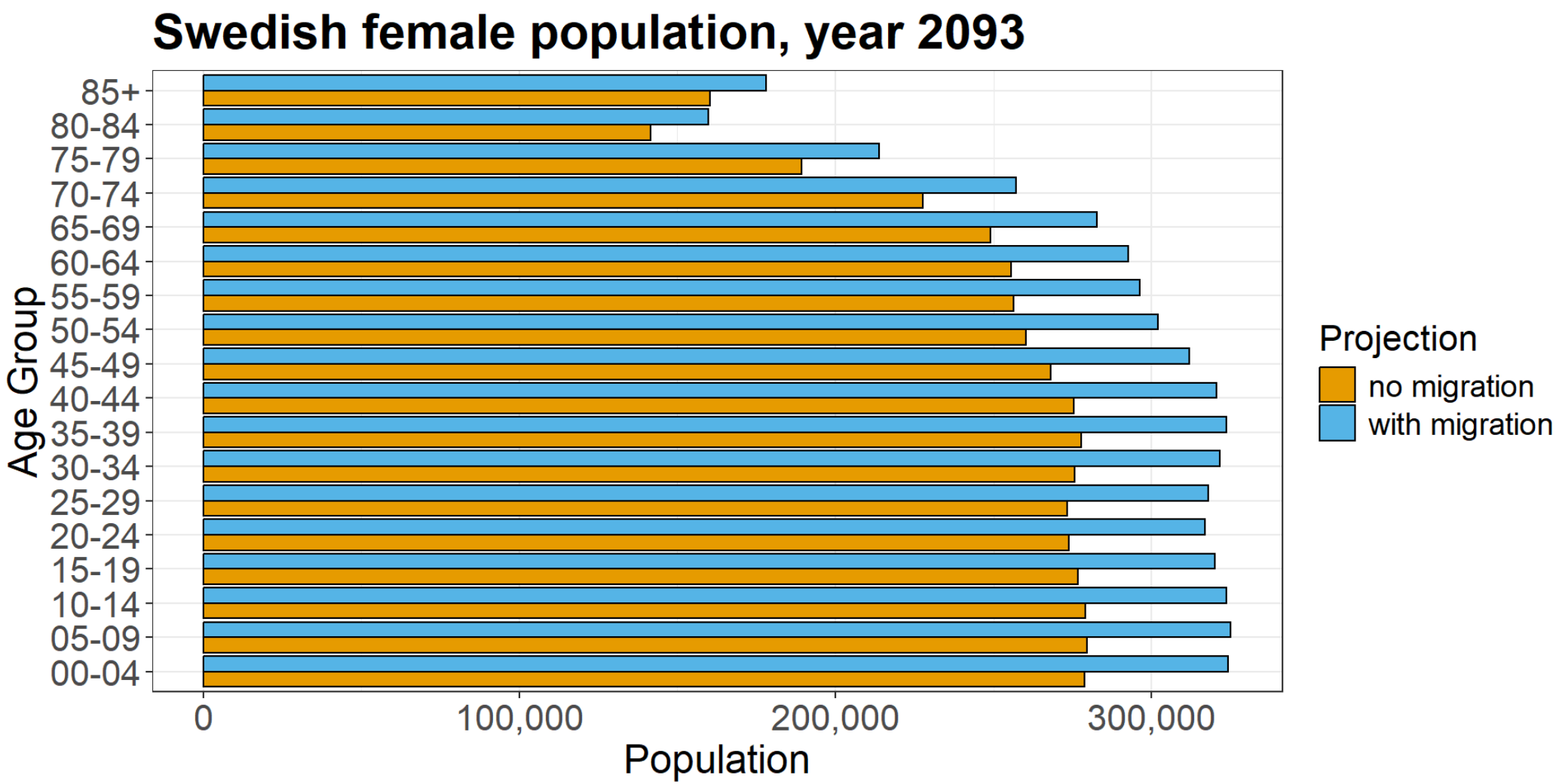
Reminder:

$$\boldsymbol{N}(t+5) = \boldsymbol{L}[t, t+5]\left(\boldsymbol{N}(t) + \frac{1}{2}\boldsymbol{I}(t)\right) + \frac{1}{2}\boldsymbol{I}(t)$$

One possible solution

```r
## derive net female migrants by age (RC schedule)
I <- 2.5e4
Ix <- I*mx
## create a function for projecting female population
## in the long term WITH net migration
pop.proj.F.mig.fun <- function(age,agegroup,sFx,bFx,
                               NFx,Ix,
                               n=1){
  ## dimension of data
  m <- length(age)
  ## create an empty Leslie matrix
  L <- matrix(0,nrow = m,ncol = m)
  ## fill up L
  L[1,] <- bFx
  diag(L[-1,]) <- sFx
  L[m,m] <- sFx[m-1]
  ## create a matrix of population vectors
  N <- matrix(0,nrow = m,ncol = n + 1)
  ## fill first column with the starting population
```

One possible solution

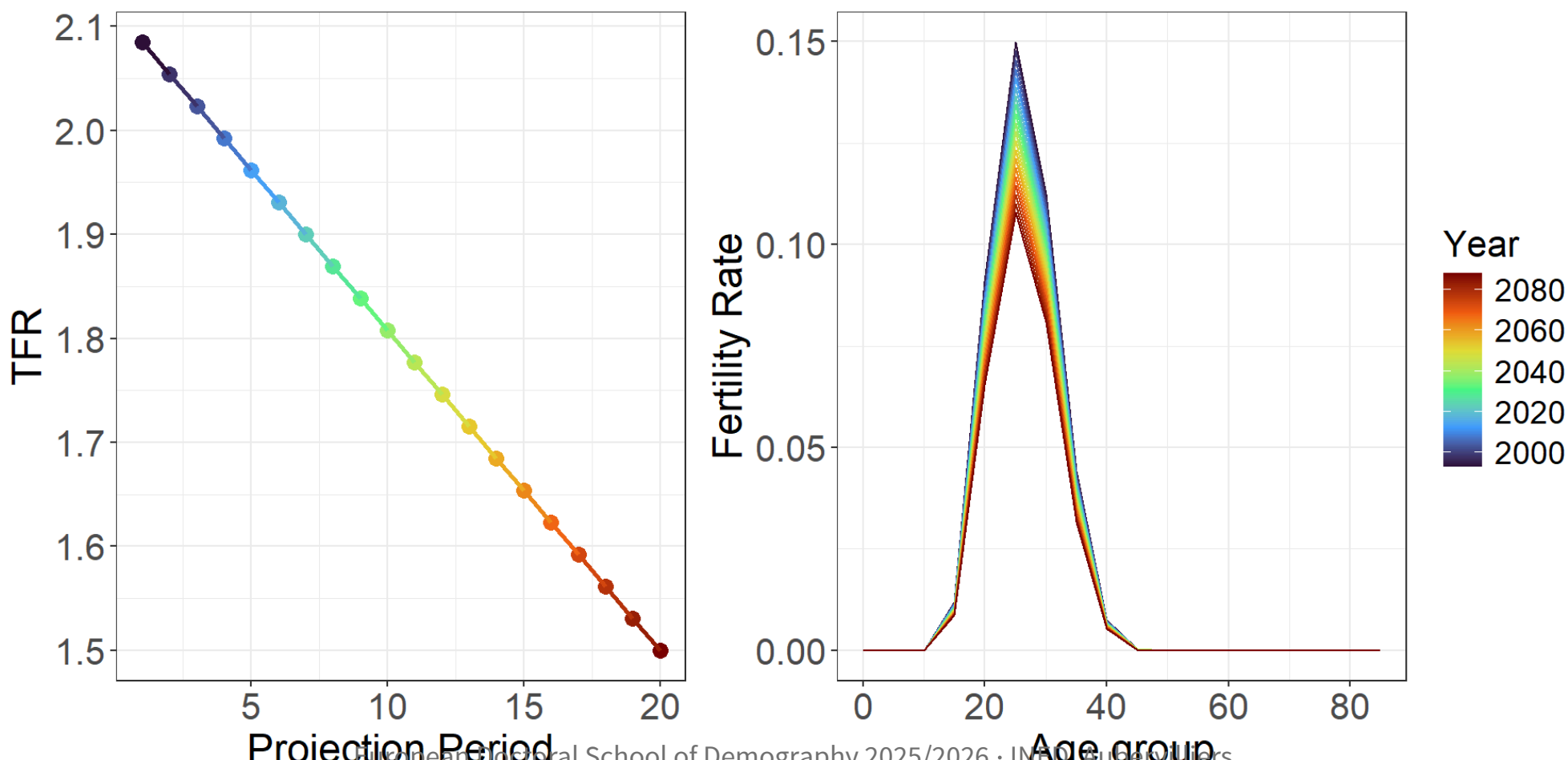


CCM: Time-varying assumptions

- it is unlikely that demographic components will not change over time
- the CCM can accomodate time-varying assumptions on some (or all) demographic components
- this extension can be used to produce more “realistic” scenarios of population projections
- this can be achieved by including time-specific entries in the Leslie matrix

An Example: Fertility decline

Let us assume that the Total Fertility Rate (TFR), $TFR = 5 \sum_x Fx$, declines from its current value in our data (= 2.085) to 1.5 at the end of the projection period



Time-varying assumptions: Exercise

Exercise

Project the female population for 20 periods ahead assuming the above pattern of TFR decline (from 2.085 to 1.5). Compare your results with the projection with unchanged fertility

Reminder: time-varying fertility rates enter the Leslie matrix via

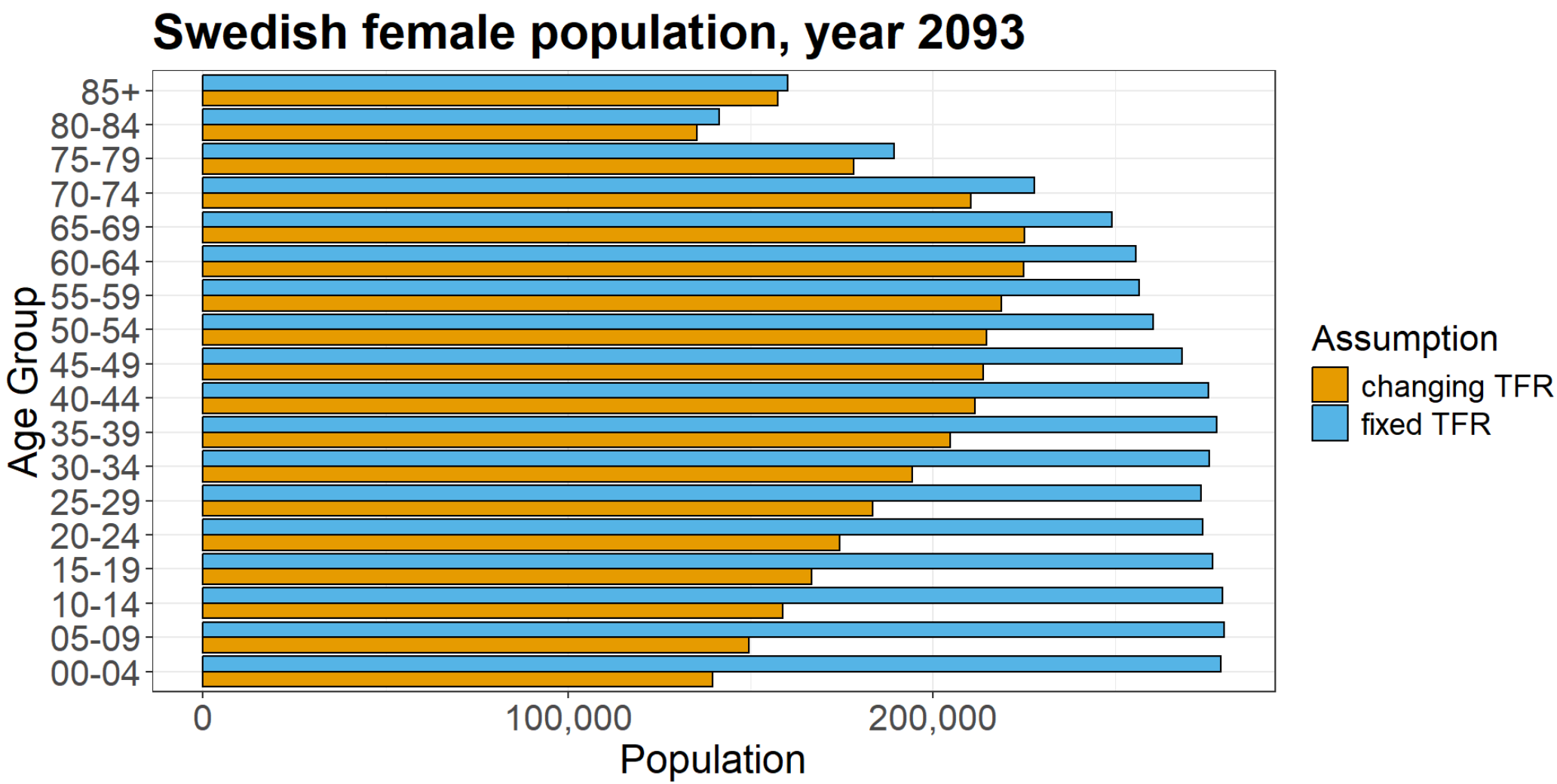
$$b_x^F(t) = \frac{1}{1 + SRB} \frac{L_0^F}{2\ell_0} \left(F_x(t) + s_x^F \, F_{x+5}(t) \right)$$

One possible solution

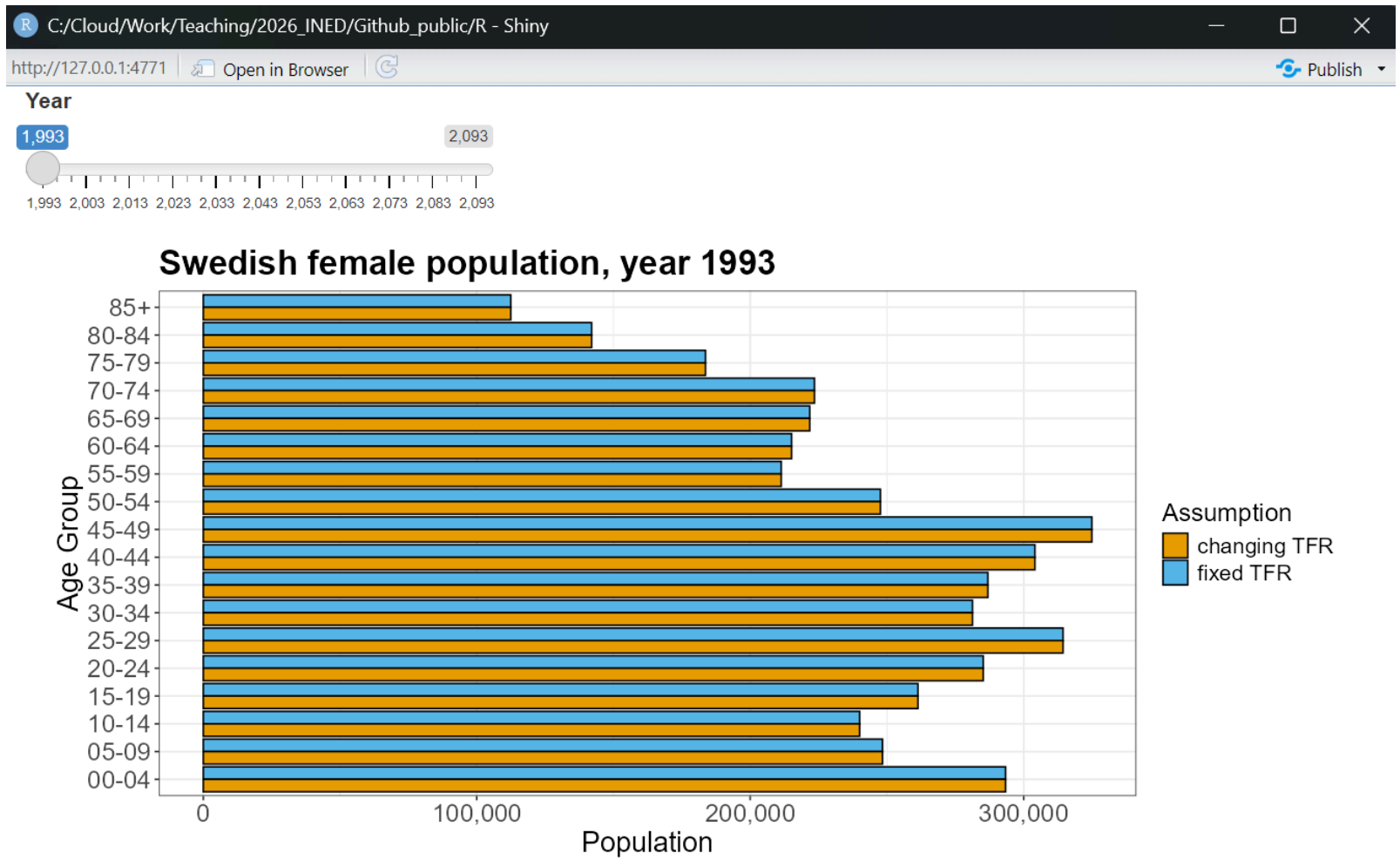
```r
## computing TFR
fx <- df.projection$Fx
tfr1 <- sum(5*fx)
m <- length(fx)
n <- 20
## adjusting TFR
TFR <- seq(tfr1,1.5,length=n)
plot(1:n,TFR)
## adjusting fertility rates
FX <- matrix(fx,m,n)
for (i in 1:n){
  FX[,i] <- FX[,i]/tfr1 * TFR[i]
}
5*apply(FX,2,sum)

## useful objects
srb <- 1.05
fact.srb <- 1/(1+srb)
LE0 <- df.projection$LEx[1]
```

One possible solution



One possible solution



Results in a shiny app

Stable Populations

(see other slides)

Some final remarks

- we have seen several extensions of the cohort component model in matrix formulation
- age interval can be of any length (1y, 5y, 10y), formulas are unchanged (but the projection period is always equal to your age interval)
- results shown from the eigendecomposition of the Leslie matrix hold only:
  - in the long run, *and*
  - for a stable population (closed to migration, with constant fertility and mortality rates over time)

Day 3 assignment

Assignment

5. Following up on Exercise #4, include migration in your projection 20 period ahead. You can either use observed net migration counts (you can estimate them from the balancing equation of population growth) or the Rogers-Castro migration schedule (try to find reasonable data on total net migration counts).

6. Project your chosen population by sex for $n = 20$ periods ahead, making time-specific assumptions about one or more demographic components.

7. Derive the long-term growth rate and population distribution by eigendecomposing the Leslie matrix of your chosen population (Exercise 4), and compare your results with those that you obtain from a long projection ($n = 50$) of your population. As a second step, modify your starting population by randomly re-assigning elements on $\boldsymbol{N}(t)$ to different age groups, and project the new population in the long term. Compute the long-term growth rate and distribution of this second population, and compare it to the first

References

Abel, G.J. (2021). *migest*: Methods for the Indirect Estimation of Bilateral Migration. R package version 2.0.1. https://CRAN.R-project.org/package=migest

Rogers, A. (1990). Requiem for the net migrant. *Geographical Analysis*, **22**(4), 283-300

Rogers, A. and Castro, L.J. (1981). Model Migration Schedules. *IIASA Research Report 81*, RR-81-30